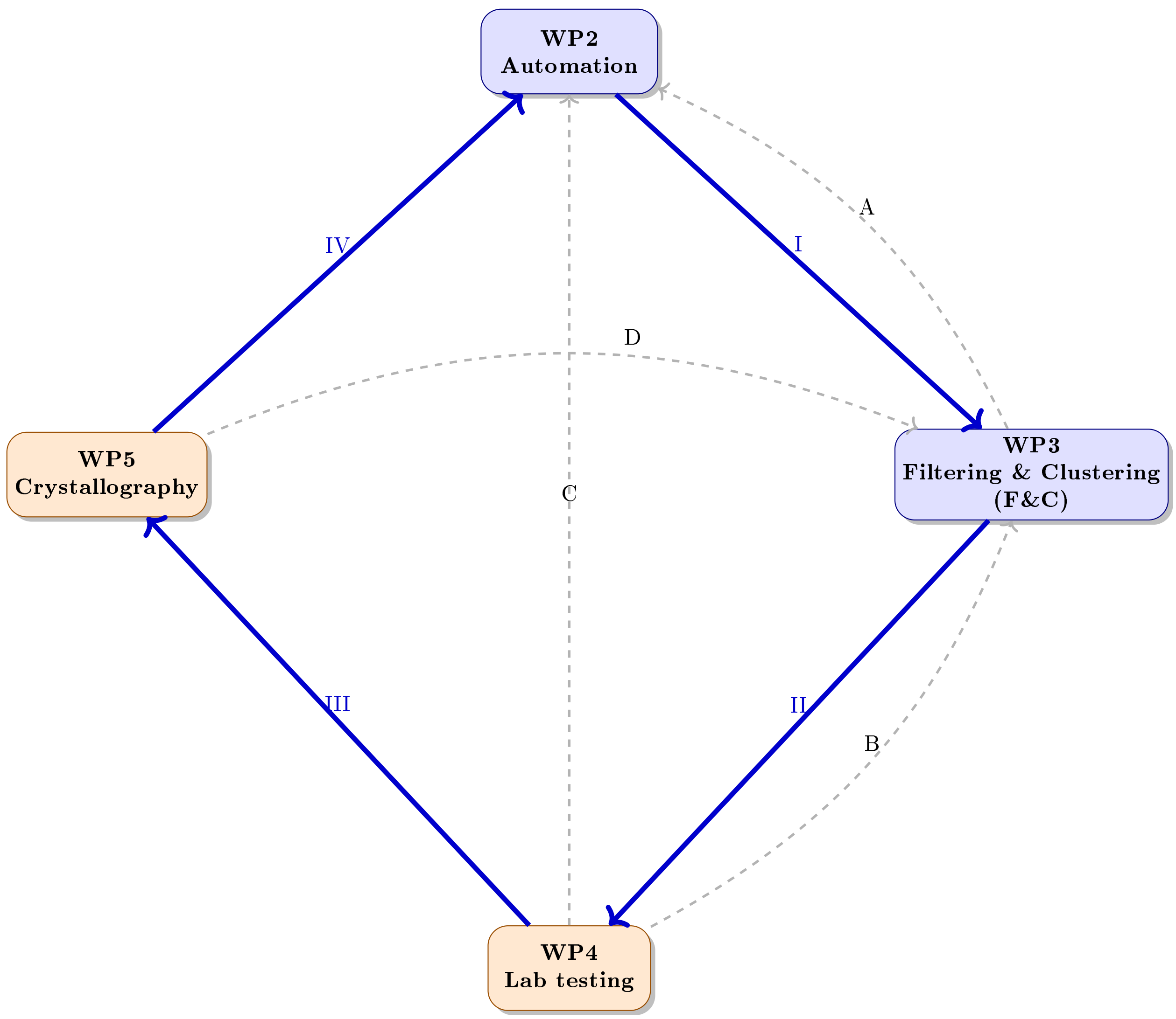
1 Workflow visualization



Primary Workflow Loop
I. Enzyme Mimic Generation (RFDiffusion)
II. Reliable Enzyme Mimic Design
III. Crystallographic Validation
IV. Structure-Guided Workflow Adaptation

Feedback and Refinement Loops
A. Workflow adjustment
B. Filtering & Clustering refinement
C. Lab-driven workflow adaptation
D. Structure-guided refinement

Explanations. WP2 automation consists of several steps, roughly speaking we have:

– Enzyme structure generation based on the use of `RFDiffusion`
– Deep Learning utilization (`ProteinMPNN`) to predict sequences that suite realistically to these structures (multiple of them)
– The generated sequence serve as input of `AlphaFokd` which generate a set of structures for these sequences.

In order to balance intensification and diversification of the search towards generation of Enzyme mimics (EMs) with a preserved catalytic functionality of the reference enzyme, two complementary techniques are employed: **filtering** and **clustering** techniques.

Filtering techniques utilize different scores to measure structural distance between generated enzymes (their structures) from the reference enzyme and filter out those based on biologically-motivated threshold value. One of the used measure is the well-known the rooted mean square deviation (RMSD). In this way, we try to narrow the search towards those with potentially more functional enzymes with an inherited catalytic property.

Clustering techniques serve to diversify the search, thus spread the search uniformly in the space of EMs. In order to be able to utilize the classical clustering such as k-means or agglomerative clustering, we map the search of EMs into the structural space. As EMs are represented as graphs (with a threshold value), examples of the graphs of several EMs from our preliminary experimental analysis are given in Figures 1–2.

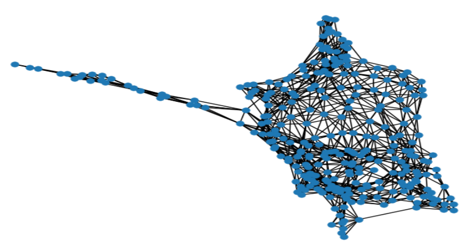


Fig. 1. `enzyme_mimicB1-501-379-00EN379a_B1-501-379-00-enzyme_mimicB1-501-379-00EN379a-ranked_0.png`

1.1 Protein Structure Graph Construction and Feature Extraction

To enable graph-based analysis of enzyme structures, each Protein Data Bank (PDB) file is transformed into an undirected residue interaction graph using Bio.PDB from `BioPython` library. Let $G = (V, E)$ denote the resulting graph, where vertices represent amino acid residues and edges represent spatial proximity relationships between residues.

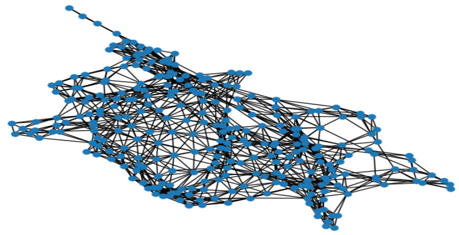


Fig. 2. `enzyme_mimicB1-501-379-03EN379i_B1-501-379-03-enzyme_mimicB1-501-379-03EN379i-ranked_0.png`

Graph Construction. For each protein structure, only residues containing a C_α atom are considered. Each residue is represented as a vertex

$$v_i \in V,$$

where v_i corresponds to the i-th amino acid residue in the protein chain. The Cartesian coordinates of the residue are taken from its C_α atom.

Two vertices v_i and v_j are connected by an undirected edge if the Euclidean distance between their C_α atoms is below a predefined threshold τ:

$$(v_i, v_j) \in E \quad \Longleftrightarrow \quad \|\mathbf{x}_i - \mathbf{x}_j\|_2 \leq \tau,$$

where $\mathbf{x}_i$ and $\mathbf{x}_j$ denote the coordinates of the corresponding C_α atoms. In our implementation, we use

$$\tau = 8\,\text{Å}.$$

This representation captures the spatial organization of the protein and encodes residue-residue contacts as graph edges.

Extracted Graph Features. For each graph, a set of topological descriptors is computed to characterize its structural properties.

- **Number of nodes**
$$n = |V|,$$
corresponding to the total number of amino acid residues.
- **Number of edges**
$$m = |E|,$$
representing the number of residue-residue contacts.
- **Graph density**
$$D = \frac{2m}{n(n-1)},$$
measuring how densely connected the graph is.

- **Average degree**
$$\bar{k} = \frac{2m}{n},$$
indicating the average number of neighboring residues.
- **Minimum and maximum degree**
$$k_{\min} = \min_{v \in V} \deg(v), \qquad k_{\max} = \max_{v \in V} \deg(v),$$
describing the least and most connected residues.
- **Average clustering coefficient**
$$C = \frac{1}{n} \sum_{v \in V} C(v),$$
where $C(v)$ denotes the local clustering coefficient of vertex v. This metric quantifies the tendency of neighboring residues to form local clusters.
- **Number of connected components**
$$c(G),$$
representing the number of disconnected subgraphs.
- **Degree centrality** For each residue,
$$DC(v) = \frac{\deg(v)}{n-1}.$$
From these values, both the average and maximum degree centrality are recorded:
$$\overline{DC} = \frac{1}{n} \sum_{v \in V} DC(v),$$
$$DC_{\max} = \max_{v \in V} DC(v).$$
- **Diameter** If the graph is connected, the diameter is computed as
$$\mathrm{diam}(G) = \max_{u,v \in V} d(u,v),$$
where $d(u,v)$ is the shortest-path distance between vertices u and v.
- **Average shortest-path length** For connected graphs,
$$L = \frac{1}{n(n-1)} \sum_{u \neq v} d(u,v),$$
measuring the average number of graph hops required to connect two residues.

These graph descriptors provide a compact representation of the protein's structural organization and can subsequently be used as input features for filtering, clustering, and machine-learning models aimed at identifying structurally diverse yet functionally plausible enzyme candidates.

0	6_enzyme_mimicB1-501EN379c	308	1591	0.033652	10.331169	4	19	0.533014	1	0.033652
1	11_enzyme_mimicB1-501EN53d	324	1632	0.031189	10.074074	4	18	0.547730	1	0.031189
2	14_enzyme_mimicB1-501EN28d	335	1659	0.029654	9.904478	4	19	0.544153	1	0.029654
3	8_enzyme_mimicB1-501EN53a	325	1599	0.030370	9.840000	2	19	0.549844	1	0.030370
4	42_enzyme_mimicB1-501EN349a	319	1607	0.031683	10.075235	3	19	0.537129	1	0.031683
5	28_enzyme_mimicB1-501EN60a	324	1613	0.030826	9.956790	2	19	0.544257	1	0.030826
6	34_enzyme_mimicB1-501EN59e	312	1540	0.031742	9.871795	2	19	0.555165	1	0.031742
7	17_enzyme_mimicB1-501EN59a	313	1579	0.032338	10.089457	2	19	0.540277	1	0.032338
8	15_enzyme_mimicB1-501EN20e	334	1622	0.029167	9.712575	4	18	0.545836	1	0.029167
9	31_enzyme_mimicB1-501EN321e	328	1655	0.030861	10.091463	2	19	0.548219	1	0.030861
10	30_enzyme_mimicB1-501EN259d	324	1582	0.030234	9.765432	3	19	0.545546	1	0.030234
11	13_enzyme_mimicB1-501EN60c	324	1633	0.031208	10.080247	2	18	0.536101	1	0.031208
12	19_enzyme_mimicB1-501EN28b	334	1648	0.029634	9.868263	4	19	0.541038	1	0.029634
13	44_enzyme_mimicB1-501EN37d	330	1666	0.030690	10.096970	2	19	0.540706	1	0.030690

Fig. 3. DataFrame of enzyme structures and their features

```
k = 2, silhouette = 0.2424
k = 3, silhouette = 0.2076
k = 4, silhouette = 0.1974
k = 5, silhouette = 0.2168
k = 6, silhouette = 0.2367
k = 7, silhouette = 0.2680
```

Fig. 4. Different k values obtained after applying k-means; somewhat suitable value selected is $k = 7$.

Clustering. Based on the best structures of each sequences generated from `AlphaFold` (50 structures, with the best score from `AlphaFold`), we have applied a k-means algorithm, with a maximum k value satisfying *silhouette* ≥ 0.25 meaning to establish a reasonable set of clusters.

Based on the consensus, several protein structures have been picked from each cluster (for example a percentage p from each of 7 clusters, or those closest to centroid of each cluster). Rest of the structures is generated random by RFDiffussion or similar ways.

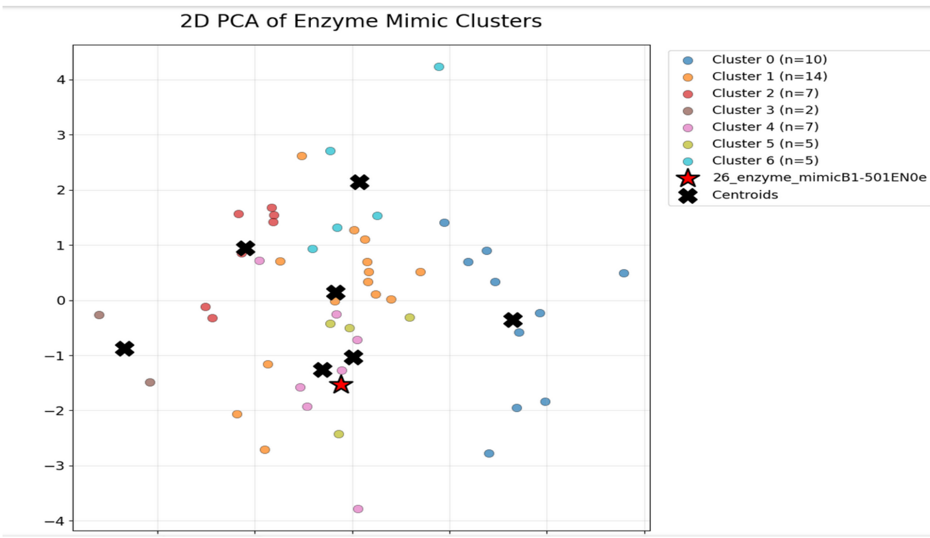


Fig. 5. 2D PCA for the best clustering of the space of EMs obtained from the workflow.